

Real Time Facial Emotion Recognition Methods using Different Machine Learning Techniques

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Abstract—Artificial intelligence has faced a rapid growth and it has made a huge change in this world. In real time the traditional algorithm has go phut to reach the human demands. Due to this great success has been gained by machine and deep learning in many different implementations such as parameter classification system, output recommendation system, different pattern recognition, etc. The thoughts, behavior and feeling of a human can be determined by his/her emotions. By utilizing the characteristics of deep learning/machine learning and different methods such as input analysis, face unlocking, etc. A facial emotions recognition system can be built and can be implemented with good accuracy. The project mainly focuses on comparison of different available deep and machine learning algorithm and find different metrics like matrix, accuracy, estimated time, validation time, etc.

Keywords—Convolution Neural Network, Machine Learning, Traditional Algorithms, Emotion Detection, Facial Emotion, Deep Learning

I. INTRODUCTION

The development of computer systems, softwares and networks and their usage are growing tremendously these days. These systems make human life much easier and they play an important role in our everyday life. A lot of importance has been assumed by facial emotions recognition system in this era since it can capture the human behavior, feelings, intentions...etc. Limited speed and less accuracy has been a major drawback for conventional system but facial emotion recognition system using deep learning has proved to be a much better one. Building a deep conventional neural network model is the main aim of this system and using that to recognize 7 different human facial emotions and using it further for applications such as customer feedback analysis, face unlocking...etc. Machine learning is one of the emerging technologies that is considered to have an impact of 95% in the next 5 years in the field of computer science. A subset of machine learning is deep learning and it uses artificial neural network, which is a computing system inspired by the human brain. A class of deep learning neural network is conventional neural network (CNN) and it uses convolution as the main mathematical operation. Different images are present inside the dataset and the system uses a 2D CNN model for the recognition of images. The proposed deep learning Convolutional neural

network is prepared not exclusively to group 7 distinct human facial feelings, however also to yield an extensively decent exactness. The presence of human emotions will play an important role in interpersonal relations. The most important application of this technology is having the ability to find out the review after the usage of any product given to them. Deep learning is the closest technology we have to real machine intelligence. In the age of big data, deep learning and machine learning are crucial for the knowledge-based applications and knowledge-based predictions. There is an endless advantage for machine learning also. This technology is responsible for the reduction of work load and time. Machine is the type of advanced technology that has the ability to evolve and still evolving based and the needs of the present generation. This helps us to improve both hardware and software simultaneously by making use of only web camera available in your laptops.

II. RELATED WORK

HariKishanKondaveeti et al. [1] proposed an emotion detection model which uses deep facial features. Pre-processing, extraction of facial features and classification of the emotions are the three important phases involve in the study of facial emotions. The comparison of deep learning techniques available in Keras system of facial emotion detection is shown using transfer learning from models like VGG16, Xception and InceptionV3 and bottleneck features are generated for our input images. Cohn-Kanade Dataset (CK+) and Japanese female facial emotion (JAFPE) dataset models are used for the performance evaluation. 83.16%, 82.15%, 77.1%, 78.11% are the accuracies obtained in this paper.

MohanaGowriet al. [2] has attempted deeplearning Convolutional Neural Network to detect facial emotions in real time. The classification of real time human facial emotions is shown and using a simple 4-layer Convolutional Neural Network (CNN) technique it is classified into one of the seven categories of emotions. From the given input images the facial features are extracted and with the help of a series of filters they are pre-processed and in order to obtain the final output image they are provided to the classifier. Improvisation of the accuracy is the main goal of this paper and also the accuracy of the architecture by engaging in different pre-handling and element extraction methods using the input images.

Navazet al. [3] proposed a facial emotion recognition system for identification and classification of different available facial emotions. The images from the datasets are compared with the real time faces using deep learning and processing methods. The method consists of three stages pre-processing, deep learning, post-processing. The accuracy of the hybrid model was found to be 93%.

Jaiswalet al. [4] proposed a neural network based approach to detect the facial emotions. If a face is detected, further processing is done to identify the shown facial emotion. CNN algorithm is used along with FER-2013 and JAFFE to optimize the loss and identify type of emotion. This shows method for using CNN and Keras by google, which is a simple and open source method available. This makes it easy to implement in real time.

A methodology for perceiving facial feelings is proposed in [5] and proceeds the Deep learning ideas utilizing the structures of Keras and TensorFlow as backend and giving a need to the Convolutional Neural Networks System. The project focuses on surveillance of emotions such as anger and fear after the classification phase. Safety of an individual is the main reason for the occurrence of fear, while coping up with the unsafe condition is the primary condition for anger according to Appraisal theory of emotions. The same neurotransmitters are used to drive the facial emotions such as fear and anger and both these facial emotions have an effect of threat perceptions. When the higher probability range is passed the user is notified about these emotions and thus completing the surveillance task. A large and publicly available dataset which is FER2013 is used for this task. Since it is a challenging dataset the performance result obtained is 64.54%.

Use of AI methods which require less assets to create similar outcomes for feeling expectation is proposed in [6]. Clinically perceived facial activity coding framework (FACS) is the premise of the feeling expectation procedure accessible and activity units (AUs) are contributed for the anticipated facial feelings for additional examination. Many different available health issues related to facial muscle movements can be analyzed due to this since this analysis acts as a main resource and complements and strengthens it. The facial feeling acknowledgment is executed by utilizing CNN network based model with LeNet design in [7] and is utilized to work on the forecast of results. A dataset, which is downloaded from Kaggle web assets, website is utilized here. Anger, happy, neutral, fear, disgust, sad and surprise are the available emotions in this datasets. Gender classification is also involved in this system along with facial emotions classifications. In social media and social networking websites automatic gender recognition is highly relevant. Through face emotion detection using Convolution Neural Network (CNN), gender and facial emotion recognition are explored in this proposed system.

Using a deep learning approach using a Convolutional Neural Network (CNN) algorithm to explore the recognition of human facial expressions is proposed in [8]. A labeled dataset containing around 32,299 images with multiple facial emotions is used by this system for testing and training. A face detection sub-system with noise removal mechanism, which also includes feature extraction technique are involved in the pre-training phase. Identification of seven emotions of the Facial Action Coding System (FACS) is made possible

by this generated classification model. An accuracy of 79.8% for the recognition of all seven basic human facial emotions has been obtained as the result of our work, without the application of optimization techniques. With the highlights separated from the Speeded Up Robust Feature (SURF) and utilizing the Convolution Neural Network (CNN), a programmed face feeling grouping framework has been proposed in paper [9]. The proposed model which supports following human feelings with looks had the option to accomplish a precision of 92%. To consequently distinguish worker feelings progressively utilizing AI a Real time Employee Emotion Detection System (RtEED) has been proposed in paper [10]. Through messages identified emotions will be intimated to respective employee which helps the employer in checking the well-being of employees. This can improve their concentration level towards work and adopt to the healthier life style and adopt much productive workstyles thereby aiding the employees in making better decisions. To train this machine learning model CMU Multi-PIE Face Data is used. To capture facial expressions of an employee in real time each employee must be equipped with a webcam. Both LBP and facial landmark features to detect emotions are fused in paper [10] and for the classification of different emotions Support Vector Machine (SVM) is used. Cohn-Kanade dataset is obtained, analyzed and exhibited and the performance has been displayed. We obtain an accuracy of 86.7%, with this proposed method.

III. PREPROCESSING METHOD –IMAGE PROCESSING

From the given, the objects must be identified. The image processing techniques would probably be the start of this process and they include noise removal, low-level feature extraction to locate certain texture and lines and regions. Interpreting these shapes as single object is the clever way like cars on road, boxes on conveyor belt or cancerous cells under a microscopic view. An object can appear very different when viewed from different angles or under different lighting, this is the one main artificial intelligence problem. Which feature belongs to what object and which is a background or shadows is also another problem. In order to approach basic human performance of a human visual system in performing these available tasks skillful programming and lots of processing power is needed by a computer system since they do not perform these tasks unconsciously. Through several possible techniques manipulating given data in the form of an image is required slide, television screens, or movie screen are the most common ways in which these given images are represented and two-dimensional array of brightness values is the most common way in which they are interpreted.

IV. ALGORITHMS

A. CONVOLUTIONAL NEURAL NETWORK

To reduce the number of parameter and adapt the network architecture specifically to the required task a bionic Convolution Neural Network is used in the project. Based on their functionalities the set of layers present in the convolution neural network can be grouped. They are follows.

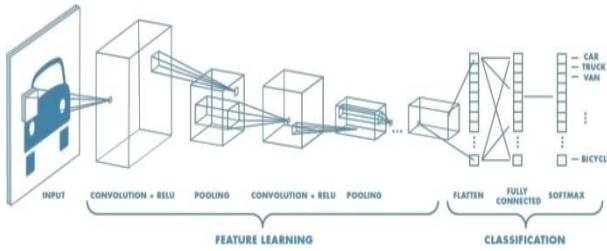


Fig.1.Representation of CNN

1) CONVOLUTION LAYER

2D convolution on the data sources is the principal thought of this undertaking. “Integrated” across “channels” are the “dot products” between the given weights and inputs. Across receptive fields filter weights are shared. The same number of layers as the input volume channels are possessed by the filter and same “depth” as the number of filters are possessed by the output volume. Fig.2 shows the working of convolution layer.

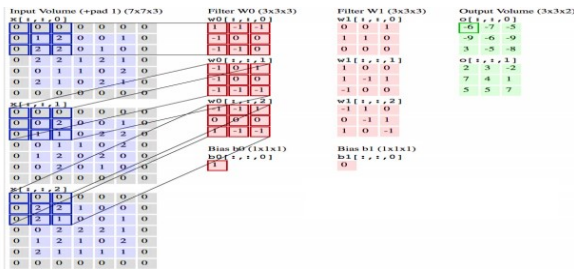


Fig.2.Convolution Layer

2) ACTIVATION LAYER

Nonlinearity of the given network are increased with out affecting receptive fields of convolution layers. For results in faster training ReLU is preferred. The vanishing gradient problem, are resolved by leaky ReLU which combines non-linearity and rectification layer. A special kind of activation layer-Soft max is used usually at the end FC layer output. Can be considered as a fancy normalizer. A discrete probability distribution vector is produced. When it is combined with cross-entropy loss it becomes very convenient. The representation of the activation layer is shown in (1). The right hand side of the equation denotes the soft max layer.

$$P(y = j | \mathbf{x}) = \frac{e^{\mathbf{x}^T \mathbf{w}_j}}{\sum_{k=1}^K e^{\mathbf{x}^T \mathbf{w}_k}} \quad (1)$$

3) POOLING LAYER

Activation maps are provided by these convolution layers. Nonlinear sampling on activation maps is applied by the pooling-layer. Aggressiveness (discarding information) is the nature of pooling. Using smaller filter size and abandon the pooling is the new trend. An example of the working of the pooling layer is shown in fig. 3.

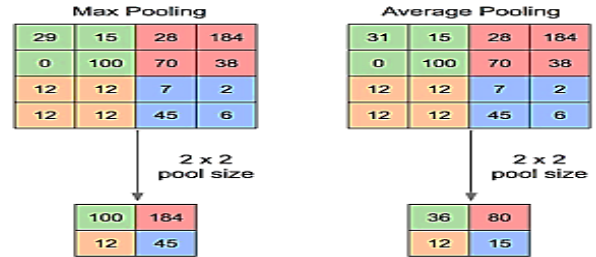


Fig.3. Pooling Layer

4) FULLY CONVOLUTED (FC) LAYER

Neural network regularly uses it. It can be considered as a final learning phase. Extracted visuals are mapped to the desired outputs. It usually is adaptive to the given classification and different encoding tasks given. Vector is the common output form which is then passed through the SoftMax and is then used to represent a confidence of classification. Different bottle necks is the ways the different received are used. It is mostly used for non-linear combinations.

B. K-NEAREST NEIGHBOR ALGORITHM

A non-parametric supervised learning algorithm such as KNN is used to classify the given data point and assign it to a given category with the help of a training set. To predict the classification of a newly appearing sample point the given data points are separated into several different classes and is given to different datasets. Fig. 4 shows the flowchart of the KNN algorithm.

CLASSIFICATION STEPS

- 1) Training Phase: Using the available training instances a model is constructed. Relationships between predictors and target are found by classification algorithm and relationships are summarized in a model.
- 2) Testing Phase: On a given test sample the model is tested with known class labels but it is not used for training the available model.
- 3) Usage Phase: For the classification of new data the model is used but class labels are unknown.

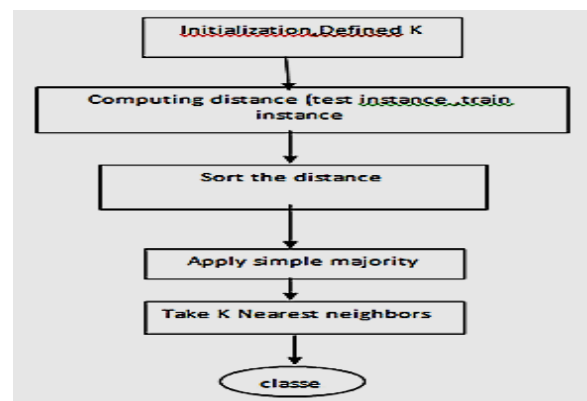


Fig. 4. KNN algorithm

C. NAÏVE BAYES

Fundamental prologue to the Bayes hypothesis is given. It was named after Thomas Bayes from the 1700s. The conditional probability, is the principle on which the Naïve Bayes classifier works. The simple available concepts of probability is used for explanation. Tossing two coins is considered as the given sample. If two of the coins are tossed and the different possibilities are inspected, {HH, HT, TH, TT} would be the sample space. The math on this probability is calculated by denoting it as P. In this event a portion of the probabilities would be as per the following: of getting two heads = $\frac{1}{4}$, of no less than one tail = $\frac{3}{4}$, of the subsequent coin being head and the main coin is tail = $\frac{1}{2}$, of getting two heads and the primary coin is a head = $\frac{1}{2}$. The restrictive likelihood of the given occasion A is given by the Bayes Theorem if and only if the event B have been occurred. The primary coin throw will be B and the subsequent coin throw will be A in the event that this case is valid. Disarray could happen on the grounds that we have switched the request for them and we go from B to A rather than A to B. As indicated by Bayes Theorem:

$$\text{Probability} = P(A/B) = [P(B/A)*P(A)] / P(B) \quad (2)$$

V. EXPERIMENTAL RESULTS

A) CONVOLUTION NEURAL NETWORK

Data pre-processing using various image processing techniques has to be done in the initial stage before the implementation. Once the dataset is processed it is Passed through pooling layer where it's dimensions are reduced for future maps. Then it is passed through the flatten layer which reduces the multidimensional input into one dimension. Then it is passed through fully connected layer where the inputs are connected every single activation unit of the other layers. Finally a fully trained model is achieved.

Image acquisition using camera takes place after the training phase. Then face detection using CNN algorithm process starts. This is followed by extracting features of the real time face present. Now the CNN model compares the real time face with the datasets. Now finding a match for the image process takes place. If the Facial expression is detected then the result in real time along with gender approximation will be given based on the datasets used along with confusion matrix and accuracy. The real time outputs of the various emotions using CNN algorithm are shown in Fig. 5.

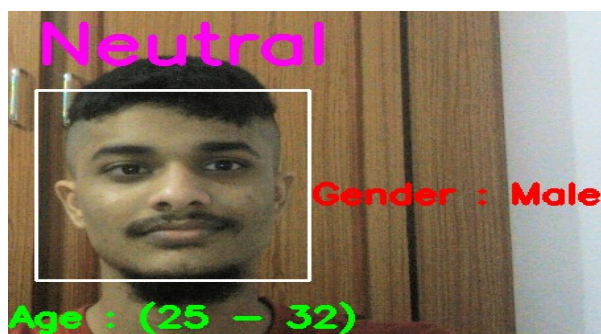


Fig.5a.Neutral face emotion

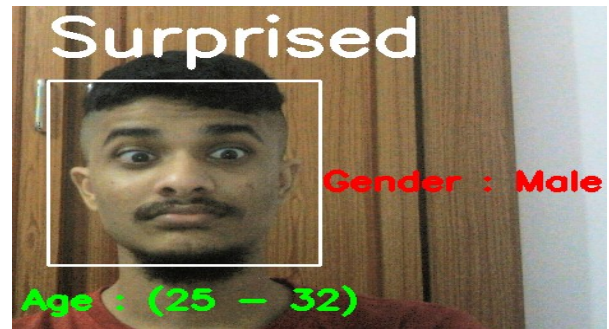


Fig.5b.Surprised face emotion

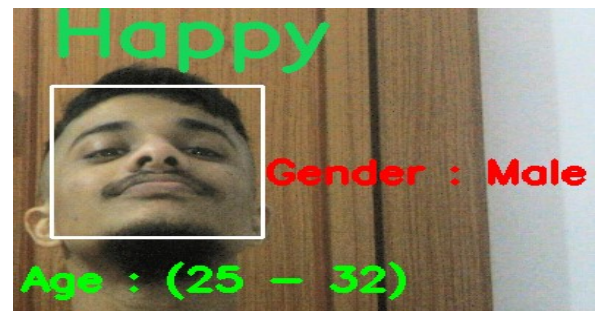


Fig.5c.Happy face emotion



Fig.5d.Fearful face emotion

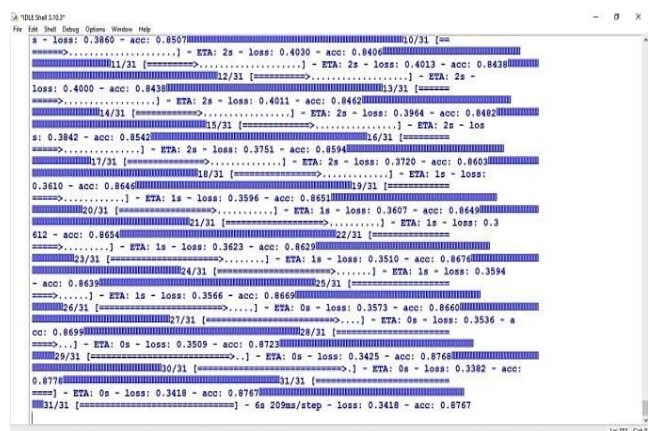


Fig.5e.Accuracy of CNN

B) KNN ALGORITHM

The process of KNN implementation begins with importing necessary packages from the open source libraries. Segregation of colors for the given dataset takes place this step is also a pre-processing step. Creation of edited dataset takes place after the previous step and they will be considered as different classes. Now the different datasets will be called and the images from the real time will be read using a web camera. Now the facial detection file created for this file will be activated. Comparison with the emotions present in the classes will take place. On successful comparison results will be displayed in real time along with training and testing values and confusion matrix. The real time outputs of the various emotions using KNN algorithm are shown in Fig. 7.

```
<generator object list_files at 0x000001E798915C10>
0.7195121951219512
[[31 0 0 1 7 2 1]
 [ 0 10 0 0 0 0 0]
 [ 1 1 31 1 7 0 4]
 [ 0 0 0 11 2 2 1]
 [ 1 1 6 1 33 1 5]
 [ 0 0 2 2 2 10 0]
 [ 0 3 5 4 3 3 5]]
```

Fig.6. Accuracy and Confusion matrix of KNN



Fig.7a. Happy face emotion

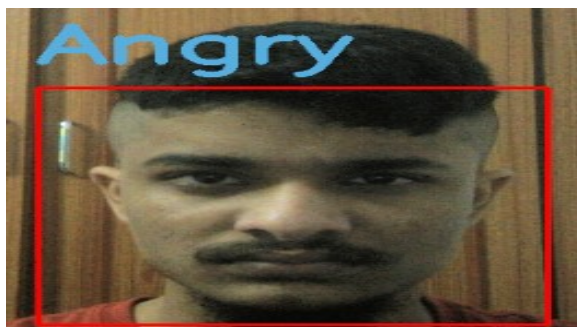


Fig.7b. Angry face emotion



Fig.7c. Sad face emotion

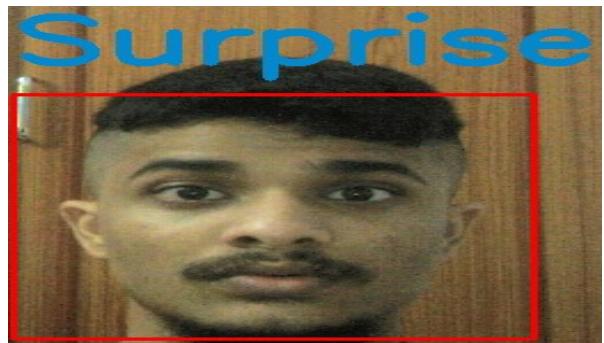


Fig.7d. Surprise face emotion

```
accuracy_score: 0.3902439024390244
confusion_matrix: [[14 1 3 2 2 0 1]
 [ 2 5 1 2 0 2 0]
 [ 7 5 21 1 6 1 4]
 [ 4 2 2 2 4 1 2]
 [10 4 11 7 21 0 8]
 [ 5 1 4 2 7 5 2]
 [ 4 3 11 3 12 1 28]]
```

Fig. 8. Confusion matrix

C) NAÏVE BAYES ALGORITHM

The process for face emotion recognition using naïve bayes algorithm begins. Each attribute A is taken. An attribute is nothing but a predictor. Traverse attribute list for A at examined node is found. Naïve bayes works on the basis of probability so find probability using value of A to be in class which is nothing but the dataset. Updates class for A. Checks whether all values are present (A). Checks whether there is any attribute match. If not it moves to next attribute. The real time results will be displayed along with metrics such as the confusion matrix as shown in Fig. 9.

```
<generator object list_files at 0x00000203941A7970>
0.25203252032520324
[[ 4  8  4  0 16  0  3]
 [ 0 15  0  0  3  0  0]
 [ 9  5  4  0 25  0  9]
 [ 3  2  1  0  8  0  4]
 [ 3  2  3  0 32  0  9]
 [ 3  2  0  0 15  0  3]
 [ 2  4  0  0 38  0  7]]
```

Fig.9. Confusion matrix

TABLE I. RESULTS OF DIFFERNET EMOTIONS

Emotions	Training Accuracy (%)	Validation Accuracy
Anger	96	0.4000
Contempt	94	0.512
Fear	92	0.432
Happy	98	0.5000
Sad	97	0.3345
Surprise	92	0.6645

TABLE II. TRAINING AND VALIDATION LOSS

S.No	Leaf Image	Estimation Time (Mins/Step)	Training Loss	Validation Loss
1.		1 ½ mins/Step	0.107	9.2967
2.		2 ½ mins/Step	0.916	9.3558
3.		2 mins/Step	0.052	7.5297
4.		3 mins/Step	0.061	6.1533
5		4 mins/step	0.071	7.2225

6		5 mins/step	0.045	7.6295
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VI. CONCLUSION

The Convolutional Neural Network model, the K-Nearest Neighbor model and Naïve Bayes model was implemented in Tensor Flow backend system to classify the facial emotions. The equivalent was executed for continuous information utilizing python IDE and OpenCV. The most normally utilized enhancers and enactment capabilities were examined. It very well may be derived that the proposed capability advances the framework with more prominent precision of 85%. It is construed that particularity and responsiveness are over 80%. This paper proposes a two-layer convolution network model for facial feeling acknowledgment. The model groups 5 distinct facial feelings from the picture Dataset. The model has similar preparation precision and approval exactness which convey that the model is throwing a tantrum and is summed up to the information.

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